CHAPTER 6

MEASURES OF CENTRAL TENDENCY

CHAPTER ESSENCE AT A GLANCE:
Master mathematical and positional averages: Arithmetic Mean, Combined Mean, Median, Quartiles, Mode, Geometric Mean, Harmonic Mean, and empirical relationships.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Ideal Average Criteria:** An ideal measure must be rigidly defined, easy to comprehend and calculate, based on all observations, capable of further algebraic treatment, and resilient to sampling fluctuations.
- **Arithmetic Mean Mathematical Invariants:** The sum of deviations from mean is always zero; the sum of squared deviations from mean is minimum; combined mean connects multiple sub-groups.
- **Positional Averages: Median and Quartiles:** Median divides an ordered array into two equal halves; Quartiles partition data into four equal quarters (Q1 = 25th percentile, Q3 = 75th percentile).
- **Mode and Empirical Relationship:** Mode represents the value of maximum frequency density; in moderately skewed unimodal distributions, Mode = 3 Median - 2 Mean.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{Arithmetic Mean} \Longleftrightarrow \bar{x} = \frac{\sum f_i x_i}{N} = A + \left(\frac{\sum f_i d_i'}{N}\right) \times h$$

Step-Deviation Computational Formula for Grouped Mean

$$\text{Combined Mean} \Longleftrightarrow \bar{x}_{12} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

Weighted Pooling of Sub-Sample Averages

$$\text{Grouped Median} \Longleftrightarrow M = L + \left(\frac{\frac{N}{2} - \text{cf}}{f}\right) \times h$$

Interpolation Formula for 50th Percentile

$$\text{Grouped Mode} \Longleftrightarrow Z = L + \left(\frac{f_1 - f_0}{2f_1 - f_0 - f_2}\right) \times h$$

Modal Class Peak Density Interpolation

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Always sort raw data in ascending order before calculating Median or Quartiles; skipping sorting leads to catastrophic errors.
- **Alert:** In continuous median and quartile interpolation, cf represents the cumulative frequency of the PRECEDING class, while f is the simple frequency of the current class.
- **Alert:** For qualitative data or open-ended frequency distributions, Median is the most appropriate central measure, whereas Arithmetic Mean cannot be computed without assumptions.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State any four desirable properties of an ideal measure of central tendency. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Four Properties:** (1) It should be rigidly defined; (2) It should be based on all observations; (3) It should be easily understandable and simple to calculate; (4) It should be capable of further mathematical and algebraic treatment. **[2 Marks]**

2. The average marks of 60 boys in a class is 45 and the average marks of 40 girls is 55. Calculate the combined mean marks of all 100 students. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Given Data:** n1 = 60, Mean1 = 45; n2 = 40, Mean2 = 55. Total N = 60 + 40 = 100. **[1 Mark]**

• **Formula:** Combined Mean = (n1 * Mean1 + n2 * Mean2) divided by (n1 + n2). **[1 Mark]**

• **Calculation:** Combined Mean = (60 * 45 + 40 * 55) divided by 100 = (2,700 + 2,200) divided by 100 = 4,900 divided by 100 = 49 marks. **[1 Mark]**

3. Calculate the Median marks from the following frequency distribution: Marks: 0-10 (f=5), 10-20 (f=8), 20-30 (f=12), 30-40 (f=10), 40-50 (f=5). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Cumulative Frequency):** cf = 5, 13, 25, 35, 40. Total N = 40. **[1 Mark]**

• **Step 2 (Locate Median Class):** N divided by 2 = 40 divided by 2 = 20. The cumulative frequency just greater than 20 is 25, so the Median class is 20-30. **[1 Mark]**

• **Step 3 (Parameters):** L = 20, cf = 13 (preceding class), f = 12 (median class), h = 10. **[1 Mark]**

• **Step 4 (Interpolation):** Median = L + [(N divided by 2 - cf) divided by f] * h = 20 + [(20 - 13) divided by 12] * 10 = 20 + (70 divided by 12) = 20 + 5.83 = 25.83 marks. **[1 Mark]**

4. Calculate the Mode from the following data: Class: 10-20 (f=4), 20-30 (f=10), 30-40 (f=25), 40-50 (f=15), 50-60 (f=6). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Identify Modal Class):** The maximum frequency is 25, which corresponds to the class 30-40. Modal Class = 30-40. **[1 Mark]**

• **Step 2 (Parameters):** L = 30, f1 = 25 (modal frequency), f0 = 10 (preceding frequency), f2 = 15 (succeeding frequency), h = 10. **[1 Mark]**

• **Step 3 (Formula):** Mode = L + [(f1 - f0) divided by (2f1 - f0 - f2)] * h. **[1 Mark]**

• **Step 4 (Calculation):** Mode = 30 + [(25 - 10) divided by (2*25 - 10 - 15)] * 10 = 30 + [15 divided by (50 - 25)] * 10 = 30 + (15 divided by 25) * 10 = 30 + 6 = 36. **[1 Mark]**

5. State two mathematical properties of Arithmetic Mean. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Property 1:** The algebraic sum of deviations of individual observations from their arithmetic mean is always zero: sum of $(x_i - x_{bar}) = 0$. **[1 Mark]**

• **Property 2:** The sum of squared deviations of observations from their arithmetic mean is minimum compared to deviations from any other arbitrary value. **[1 Mark]**

6. In a moderately asymmetric distribution, the Mean is 30 and the Median is 28. Find the value of Mode using the empirical relationship. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Empirical Formula:** Mode = 3 Median - 2 Mean. **[1 Mark]**

• **Calculation:** Mode = 3 * 28 - 2 * 30 = 84 - 60 = 24. **[1 Mark]**